META-ANALYSIS

«Forest Plot Interpretation, Model Selection & Outlier Detection»

Scenario-Based Task (Using Real Data)

Name:Hamid Reza Ghahramani

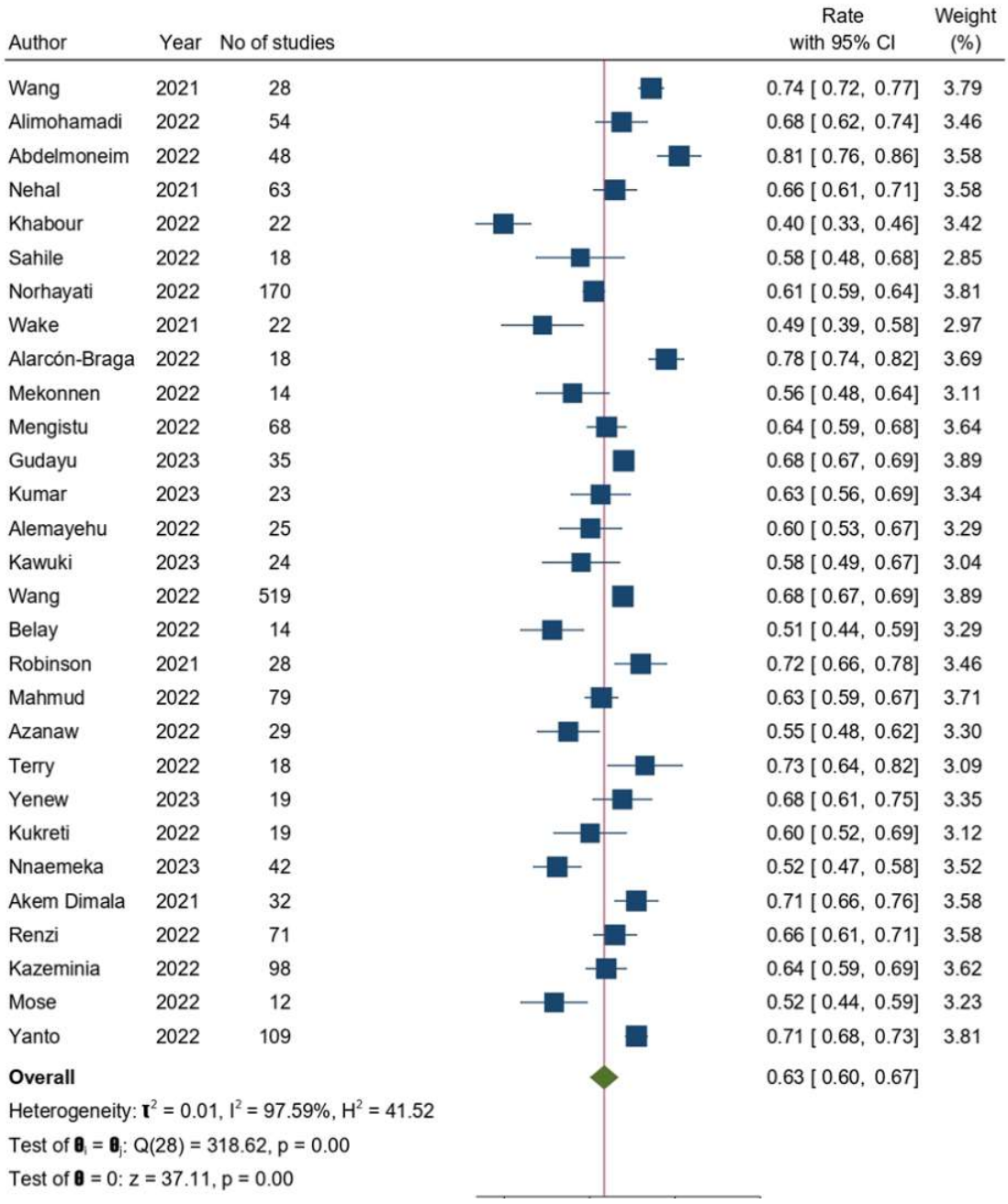
Source article

Al Rahbeni T, Satapathy P, Itumalla R, et al. **COVID-19 Vaccine Hesitancy: Umbrella Review of Systematic Reviews and Meta-Analysis.** JMIR Public Health and Surveillance. 2024;10:e54769. doi:10.2196/54769

There is no need for you to have the full version of the article for this task. But it is available in PubMed if you like to read it.

Figure 2: Forest Plot

Figure 2. Forest plot depicting the pooled acceptance rates for COVID-19 vaccines in the general population. REML: restricted maximum likelihood.



Section A — Reading the Forest Plot

1. Explain what each of the following represents:

(a) the square for each study:

(b) the horizontal line through the square:

(c) the size of the square:

(d) the diamond at the bottom of the plot:

The square represents the effect estimate of each study. The horizontal line represents its 95% confidence interval. The size of the square represents the study's weight in the meta-analysis. The diamond represents the pooled effect estimate across the included studies.

2. Abdelmoneim et al. (2022) reported an acceptance rate of 0.81 (95% CI 0.76–0.86). Interpret this result in words.

Abdelmoneim et al. (2022) estimated a vaccine acceptance rate of 81%, with a 95% confidence interval ranging from 76% to 86%.

3. The pooled estimate is 0.63. In this meta-analysis, what exactly does 0.63 represent?

The pooled estimate of 0.63 represents the overall estimated vaccine acceptance rate across the included studies, taking into account the weight of each study.

4. what does the "positive overall effect size" mean? Does moving to the right mean a "more beneficial treatment effect" in this analysis? Explain what moving to the right means.

A positive overall effect size does not necessarily mean a beneficial treatment effect. Its interpretation depends on the outcome and how the effect is defined. In this meta-analysis, because the outcome is vaccine acceptance, moving to the right represents a higher estimated vaccine acceptance rate.

5. Compare the pooled ES (0.63) with Khabour (0.40) and Abdelmoneim (0.81). What does the difference tell you about vaccine acceptance across the included systematic reviews?

Khabour reported an acceptance rate of 0.40 (40%), while Abdelmoneim et al. reported 0.81 (81%). The pooled estimate was 0.63 (63%), which represents the overall estimated acceptance rate across the included studies, taking study weights into account. The differences suggest variation in vaccine acceptance across the included studies.

Each square represents the effect estimate of an individual study, while the diamond represents the pooled effect estimate of the meta-analysis. The position of each square shows the effect estimate of that study, whereas the position of the diamond shows the pooled effect estimate.

Section B — Explore for the potential outliers in the forest plot. Interpret your answer.

Khabour et al. (2022) appears to be a potential outlier, with an effect estimate of 0.40 (40%), which is noticeably lower than the estimates of the other studies. Its weight is 3.42%. This difference

suggests that vaccine acceptance in this study was substantially lower than in the other included studies and may contribute to variation in the overall results.

Section C — Heterogeneity

1. The reported I^2 is 97.59%. Interpret this value.

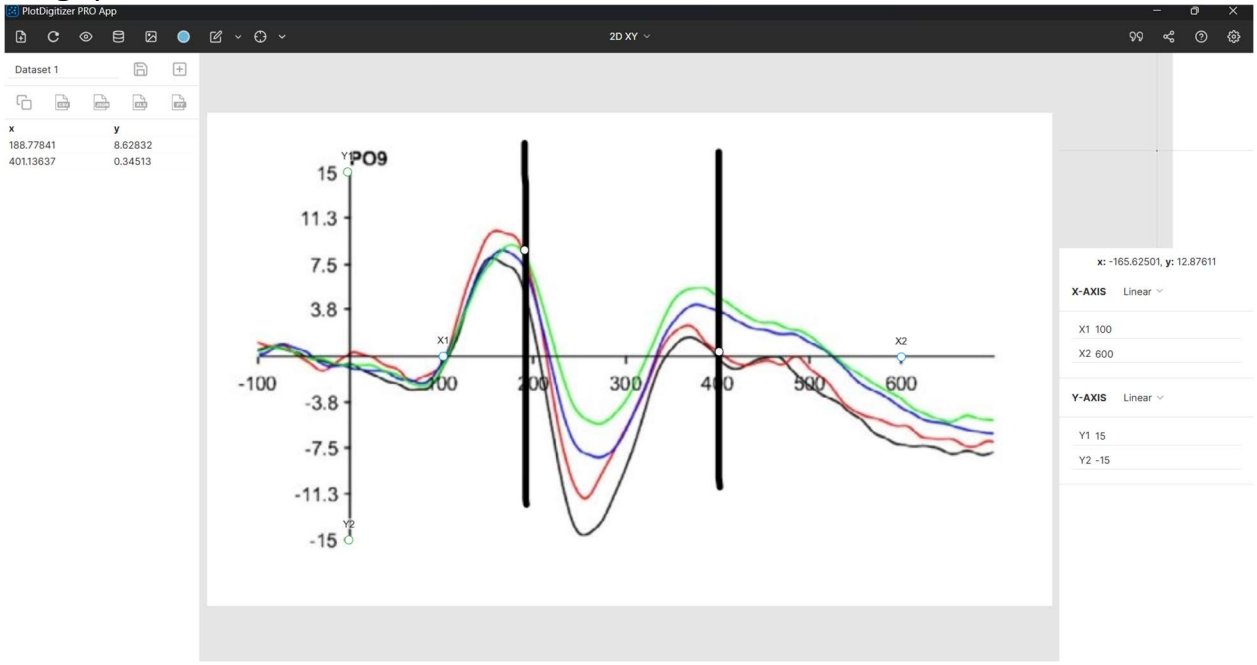
An I^2 value of 97.59% indicates very high heterogeneity among the included studies. It suggests that most of the observed variation in the effect estimates is due to real differences between the studies rather than sampling error or chance. Therefore, the effect estimates are highly dispersed, indicating substantial heterogeneity across the studies.

2. Is the very high I^2 consistent with the visual pattern in the forest plot? Give at least two features of the plot that support your answer.

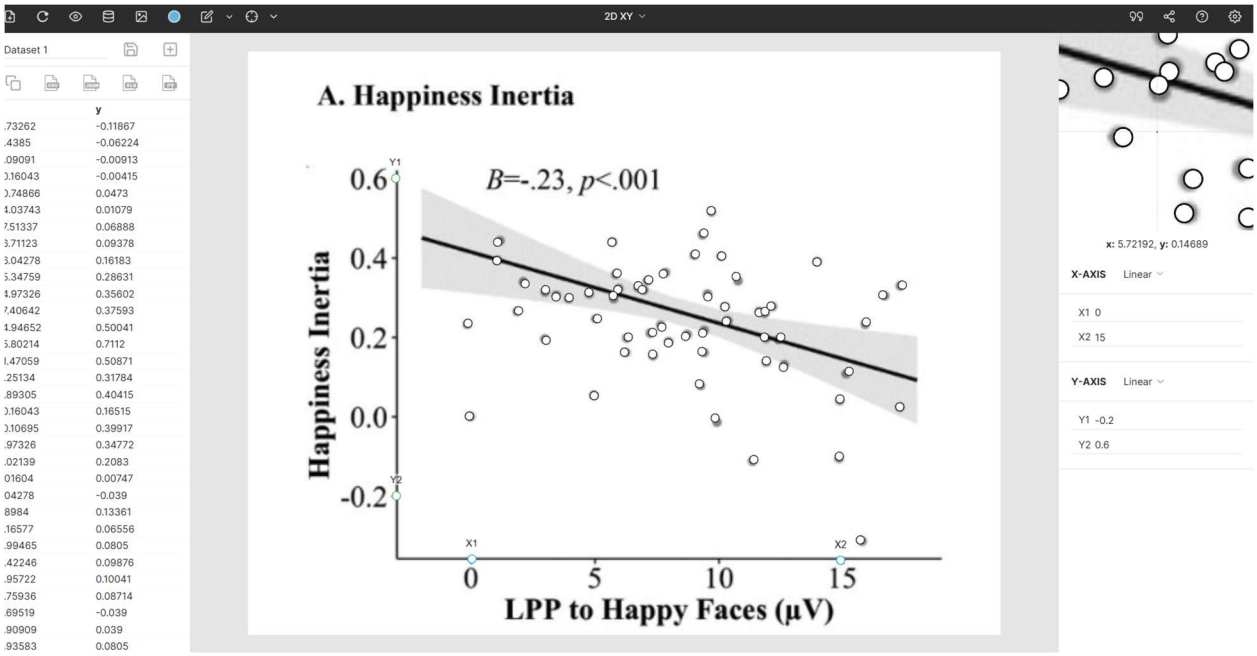
Yes, the very high I^2 is consistent with the visual pattern in the forest plot. First, the effect estimates are widely dispersed, ranging from around 40% to 70–80%, rather than being closely clustered. Second, some studies have relatively wide confidence intervals, showing differences in the precision of the estimates. Overall, the wide dispersion of the effect estimates supports the presence of very high heterogeneity.

Section D — Plot digitizer

1. Use Dataset-1 and extract the data points of **the red curve) Just the marked section in the image.)**



2. Use Dataset 2 and extract the data of **the gray points.**



3. Save both datasets (excel format) and send me the files + screenshots of the points you extracted.